

ms2: A molecular simulation tool for thermodynamic properties, release 3.0

Gábor Rutkai^a, Andreas Köster^a, Gabriela Guevara-Carrion^a, Michael Schappals^b, Colin W. Glass^c,
Martin Bernreuther^c, Amer Wafai^c, Simon Stephan^b, Maximilian Kohns^b, Stephan Deublein^b,
Martin Horsch^b, Hans Hasse^b, Jadran Vrabec^{a,*}

^a*Lehrstuhl für Thermodynamik und Energietechnik, Universität Paderborn, 33098 Paderborn, Germany*

^b*Lehrstuhl für Thermodynamik, Universität Kaiserslautern, 67653 Kaiserslautern, Germany*

^c*Höchstleistungsrechenzentrum Universität Stuttgart (HLRS), 70550 Stuttgart, Germany*

Abstract

A new version release (3.0) of the molecular simulation tool *ms2* [S. Deublein et al., Comput. Phys. Commun. 182 (2011) 2350 and C. Glass et al., Comput. Phys. Commun. 185 (2014) 3302] is presented. Version 3.0 of *ms2* features two new ensembles (*NVE* and *NpH*); the calculation of Helmholtz energy derivatives in the *NVE* ensemble, the osmotic pressure and the Quaternary Maxwell-Stefan diffusion coefficients; thermodynamic integration as a method for calculating the chemical potential; the detection of hydrogen bonding during simulation as well as the ability to sample an arbitrary number of state points in a single run.

Keywords: molecular simulation, molecular dynamics, Monte Carlo

New version program summary

Program Title: *ms2*

Operating system: Unix/Linux

Licensing provisions(please choose one): CC0 1.0/CC By 4.0/MIT/Apache-2.0/BSD 3-clause/BSD 2-clause/GPLv3/CC BY NC 3.0

Programming language: Fortran90

Has the code been vectorized or parallelized: yes (Message Passing Interface (MPI) protocol and OpenMP Scalability is up to 2000 cores)

*Corresponding author: Jadran Vrabec, Warburger Str. 100, 33098 Paderborn, Germany, Tel.: +49-5251/60-2420, Email: jadran.vrabec@upb.de

URL: <http://www.ms-2.de> ()

Distribution format: tar.gz

Classification: 7.7, 7.9, 12

Catalogue identifiers of previous versions: AEJF_v1_0 and AEJF_v2_0

Supplementary material:

Journal reference of previous versions: S. Deublein et al., Comput. Phys. Commun. 182 (2011) 2350 and C. Glass et al., Comput. Phys. Commun. 185 (2014) 3302

Does the new version supersede the previous version?: yes

Reasons for the new version: introduction of new features

Summary of revisions: two new ensembles (NVE and NpH), new properties (Helmholtz energy derivatives, chemical potential via thermodynamic integration, osmotic pressure, Quaternary Maxwell-Stefan diffusion coefficients), new functionalities (detection of hydrogen bonding, the ability to sample an arbitrary number of state points in a single run)

Nature of problem: Calculation of application oriented thermodynamic properties: vapor-liquid equilibria of pure fluids and multi-component mixtures, thermal and caloric data as well as transport properties

Solution method: Molecular dynamics, Monte Carlo, various ensembles, grand equilibrium method, Green-Kubo formalism, Lustig formalism

Typical running time: varies (typically from a couple of hours up to days) depending on the specific scenario (system size, calculated properties, number of CPU cores used)

Restrictions: typical problems addressed by *ms2* are solved by simulating systems containing typically 1000 – 4000 molecules that are modelled as rigid bodies

Documentation: documentation is provided with the installation package and is available at

1. New Ensembles (NVE and NpH) and Helmholtz energy derivatives

An ensemble is the set of all theoretically possible microscopic configurations (configurations on the molecular level) under some specific macroscopic conditions. The microcanonical ensemble (NVE) is the set of all configurations that fulfill the condition of having the same number of particles N , volume V and energy E , whereas the isobaric-isenthalpic ensemble (NpT) represents a system at constant number of particles N , pressure p and enthalpy H . Molecular dynamics (MD) simulate the time evolution of the system by numerically solving the Newton's equations of motion for each molecule. Because of the nature of this solution, the time is discretized and the method

yields microscopic configurations at discrete but consecutive time steps. The Monte Carlo (MC) method is the application of statistical mechanics to describe molecular systems. With this approach, microscopic configurations are generated and potentially accepted such a way that only relevant and physically meaningful configurations are created. The generation and acceptance of these configuration is governed by a specific probabilities [1] that are ensemble dependent and known for essentially every relevant ensemble. *ms2* For MC simulations, the NVE and NpH ensembles were implemented in *ms2* 3.0 according to Ref. [2] and [3], respectively. For MD simulations, the pressure is kept constant using Andersens barostat [4] for the NpH ensemble. Because the solution of Newton's equations of motion is approximate, the total energy of the system $E = K + U$, which consists of a kinetic K (exclusively molecular velocity dependent) and potential U (exclusively spacial coordinate dependent) contribution, is not rigorously conserved in a standard NVE run. Therefore, the velocity (translational and rotational) of every molecule is rescaled such that their sum calculated by the equipartition theorem matches $E - U$, where E is specified and U is calculated from the current configuration. For an NpH run, the solution is analogous: velocities are rescaled such that the current kinetic energy K matches $H - U - pV$, where H and p are specified, U and V are dependent on the current microscopic configuration. This extends the available ensembles to five: NVT , μVT , NVE , NpT and NpH , where μ is the chemical potential.

Table 1 contains numerical results for state points located in the dens fluid region for verification purposes. Additional to the new ensembles, the calculation of the Helmholtz energy derivatives

$$A_{nm}^r = (1/T)^n \rho^m \frac{\partial^{n+m} f^r(T, \rho)/(RT)}{\partial (1/T)^n \partial \rho^m}, \quad (1)$$

with the molar residual Helmholtz energy f^r , the temperature T , the density ρ , the molar gas constant R were also introduced for the NVE ensemble up to the order $n = 3$ and $m = 2$. It can be shown that the total $f/(RT)$ can be additively separated into an ideal $f^o/(RT)$ and a residual part $f^r/(RT)$. The ideal part by definition corresponds to the value of $f/(RT)$ when no intermolecular interactions are at work [5]. $f^o/(RT)$ consist of an exclusively temperature and an exclusively density dependent part. The former has a trivial density dependence $\ln(\rho/\rho_{\text{ref}})$, whereas the latter has a non-trivial temperature dependence. The formalism that allows for the calculation of these derivatives on the fly from a single simulation run per state point was published by Lustig [6, 7] and was already introduced for the NVT ensemble in the previous release. *ms2* 3.0 now yields these derivatives also for NVE simulations. Here, however, in contrast with NVT an simulation, there are two numerical result for each calculated derivative A_{nm} which is due to the two possible entropy definition in statistical mechanics [7]. In any case, these two different sets of results must be identical in the thermodynamic limit ($N \rightarrow \infty$). In practice, they already agree within their estimated statistical uncertainty for a simulation with around a thousand particle.

A detailed description of how to convert these derivatives to common thermodynamic properties are provided in the supplementary material of release 2.0 of Ref. [8] and in Ref. [5].

Table 1: Comparison of the ensembles implemented in *ms2* for a dense liquid state point of Methyl Fluoride (R41). Statistical uncertainties were estimated with the method of Flyvbjerg and Petersen [9]

		Temperature K	Density mol/l	Pressure MPa	Potential energy kJ/mol
MD	<i>NVT</i>	220	27	100.2 $\pm$ 0.2	-16.63 $\pm$ 0.02
	<i>NpT</i>	220	26.994 $\pm$ 0.005	100	-16.65 $\pm$ 0.04
	<i>NVE</i>	220.09 $\pm$ 0.06	27	100.1 $\pm$ 0.2	-16.63 $\pm$ 0.01
	<i>NpH</i>	219.8 $\pm$ 0.2	26.991 $\pm$ 0.004	100	-16.64 $\pm$ 0.03
MC	<i>NVT</i>	220	27	100.2 $\pm$ 0.1	-16.66 $\pm$ 0.01
	<i>NpT</i>	220	27.015 $\pm$ 0.002	100	-16.66 $\pm$ 0.02
	<i>NVE</i>	220.01 $\pm$ 0.03	27	100.2 $\pm$ 0.1	-16.65 $\pm$ 0.01
	<i>NpH</i>	220.11 $\pm$ 0.04	27.005 $\pm$ 0.002	100	-16.66 $\pm$ 0.01

2. Thermodynamic integration

The previously described method of Lustig does not allow for the direct sampling of the chemical potential thus entropic properties like A_{00} . Such an effort still requires techniques based on free energy calculation such as particle insertion and/or thermodynamic integration [1]. Widom’s particle insertion method [10] is a conceptually straightforward approach to calculate the chemical potential with low computational cost, both for pure substances and mixtures. The total chemical potential for a particle i can be separated into an ideal (o) and a residual (r) part the same way the Helmholtz energy is decomposed (cf. section 1). $\mu_i^{\text{total}}(T, \rho, x_i) = \mu_i^{\text{o}}(T) + RT \ln(N_i/V/\rho_{\text{ref}}) + \mu_i^{\text{r}}(T, \rho, x_i)$, where N_i is the number of particles with the same identity that of i , $\rho = N/V$, $x_i = N_i/N$ and ρ_{ref} is some arbitrary reference density. The part $\mu_i^{\text{total}} - \mu_i^{\text{o}}(T)$ is often referred to as the configurational chemical potential μ_i^{conf} . Widom’s method requires the frequent insertion of an additional ($i = N + 1$) molecule into the simulation volume at a random position with a random orientation. At constant temperature and constant pressure or volume the potential energy U_i of this test molecule, the interaction energy with all other ”real” N molecules, yields the configurational chemical potential according to

$$\mu_i^{\text{conf}} = \mu_i^{\text{total}} - \mu_i^{\text{o}}(T) = -k_{\text{B}}T \ln \frac{\langle V \exp(-U_i/k_{\text{B}}T) \rangle}{\langle N_i \rangle}, \quad (2)$$

where k_{B} is Boltzmann’s constant. The test molecule is removed immediately after the calculation of its potential energy U_i , thus it does not influence the properties of the system in any way. In contrast to the usual convention, the brackets $\langle \rangle$ here have a dual meaning: They stand for either *NVT* or *NpT* ensemble average as well as an integral over all the possible position and orientation of the test molecule in the system. The density of the system has a significant influence on the accuracy of this method. For state points with high density, test molecules almost always overlap with some real molecules, which leads to a potential energy $U_i \rightarrow \infty$ and thus to no contribution to Eq. 2 resulting in poor statistics and often to the complete failure of the sampling.

Thermodynamic integration is one solution to overcome the limitations of Widom’s particle insertion

method. The idea behind calculating the chemical potential using thermodynamic integration is to avoid insertion of the test particle in system A and perform it in system B for which it is not problematic or even not required at all. Since the chemical potential is a state variable, its change $\mu_A - \mu_B$ can be calculated using any arbitrary path between state A and B . This path is represented by the parameter λ . It can be shown [1] that the relation between λ and $\mu_A - \mu_B$ is

$$\mu_{A,i} - \mu_{B,i} = \int_{B(\lambda_{\min})}^{A(\lambda_{\max})} \left\langle \frac{\partial U_i(\lambda)}{\partial \lambda} \right\rangle d\lambda \quad (3)$$

for particle i . The bracket $\langle \rangle$ in this equation denotes NVT or NpT ensemble average and U_i is the potential energy of particle i that must be a part of the system the same way any other particle is. The only difference between i and regular particles is that the interaction energy $U_i(\lambda)$ is scaled between states $A(\lambda_{\max})$ and $B(\lambda_{\min})$ with λ . As long as $\partial U_i(\lambda)/\partial \lambda$ can be calculated analytically and sampled during simulation, the actual way of scaling $U_i(\lambda)$ with λ is completely arbitrary since μ is a state variable. The integration with respect to λ is evaluated numerically. Assuming that $\mu_{B,i}$ is practically zero or at least can be successfully calculated by Widoms particle insertion method at the end point B using Eq. (2), $\mu_{B,i}$ yields the configurational part of the total chemical potential. The non-linear scaling $U_i(\lambda) = \lambda^x U_i$ for $\lambda_{\min} \leq \lambda \leq 1$ with an adaptive sampling technique [11] was implemented for MC with x and $\lambda_{\min}$ being input parameters. This adaptive technique allows for the sampling of the entire range of $\lambda_{\min} \leq \lambda \leq 1$ in a single MC simulation run with arbitrary resolution for the numerical integration. Besides the standard NVT and NpT ensemble MC moves, the simulation includes changes of λ controlled by a proper MC acceptance criterion ensuring visits at each discrete λ value in the range $\lambda_{\min} \leq \lambda \leq 1$. For MD simulations, changes of λ are carried out also on the fly in a single simulation but without any acceptance criterion.

3. Quaternary Maxwell-Stefan diffusion coefficients

ms2 employs the Green-Kubo formalism based on the net velocity auto-correlation function to obtain $n \times n$ phenomenological coefficients [12]

$$L_{ij} = \frac{1}{3N} \int_0^\infty dt \left\langle \sum_{k=1}^{N_i} \mathbf{v}_{i,k}(0) \cdot \sum_{l=1}^{N_j} \mathbf{v}_{j,l}(t) \right\rangle, \quad (4)$$

in a mixture of n components. Here, N is the total number of molecules, N_i is the number of molecules of component i and $\mathbf{v}_{i,k}(t)$ denotes the center of mass velocity vector of the k -th molecule of component i at time t . Note that Eq. (4) corresponds to a reference frame in which the mass-averaged mixture velocity is zero [12].

Starting from the phenomenological coefficients L_{ij} , the elements of a $(n-1) \times (n-1)$ matrix Δ can be defined as [12]

$$\Delta_{ij} = (1 - x_i) \left(\frac{L_{ij}}{x_j} - \frac{L_{in}}{x_n} \right) - x_i \sum_{k=1 \neq i}^n \left(\frac{L_{kj}}{x_j} - \frac{L_{kn}}{x_n} \right), \quad (5)$$

so that its inverse matrix $\mathbf{B} = \mathbf{\Delta}^{-1}$ is related to the Maxwell-Stefan diffusion coefficients D_{ij} . In the case of a quaternary mixture, the Maxwell-Stefan diffusion coefficients are then calculated using [12]

$$D_{14} = \frac{1}{B_{11} + (x_2/x_1)B_{12} + (x_3/x_1)B_{13}}, \quad (6)$$

$$D_{24} = \frac{1}{B_{22} + (x_1/x_2)B_{21} + (x_3/x_2)B_{23}}, \quad (7)$$

$$D_{34} = \frac{1}{B_{33} + (x_1/x_3)B_{31} + (x_2/x_3)B_{32}}, \quad (8)$$

$$D_{12} = \frac{1}{1/D_{24} - B_{21}/x_2}, \quad (9)$$

$$D_{13} = \frac{1}{1/D_{14} - B_{13}/x_1}, \quad (10)$$

$$D_{14} = \frac{1}{1/D_{24} - B_{23}/x_2}, \quad (11)$$

$$(12)$$

where x_i is the molar fraction of component i .

4. Hydrogen bonding

There is no unambiguous characterization of a hydrogen bond between two molecules [13, 14, 15]. The H bond ‘is a structural motif and involves at least three atoms’ [16]. IUPAC defines the H bond as ‘an attractive interaction between a hydrogen atom from a molecule or molecular fragment X–H in which X is more electronegative than H, and an atom or a group of atoms in the same or a different molecule, in which there is evidence of bond formation’ [15].

Energetic [13], geometric [17], and topological [14] H-bonding criteria have been proposed in the literature. Geometric criteria, which are not overly complex, are often based on the following assumptions [18]:

- The interaction between two H-bonded sites is highly directional and short ranged.
- A donor interacts at most with a single acceptor, an acceptor can interact with multiple donors.

Accordingly, a class of geometric criteria for the evaluation of H-bonding networks in fluids is implemented in *ms2*. Thereby, the triangle between three charge sites, situated in two molecules, is evaluated to determine whether an interaction between two sites constitutes a H bond or not. A molecule acts as a H donor to another molecule, the H acceptor, if the following conditions hold [17, 18, 19, 20]:

- The distance between the donor and the acceptor is smaller than the respective threshold distance, i.e. ℓ_{AD} or ℓ_{DA} , cf. Fig. 1.
- The distance between the acceptor sites of the acceptor and donor molecules is smaller than ℓ_{AA} .
- The angle between the acceptor-donor axis and the acceptor-acceptor axis is smaller than the respective threshold angle, i.e. φ_{DAA} or φ_{AAD} , cf. Fig. 1.

Therein, ℓ_{AD} , ℓ_{DA} , ℓ_{AA} , φ_{DAA} , and φ_{AAD} are the parameters of the implemented class of H-bonding criteria; for methanol, e.g., Haughney *et al.* [17] use $\ell_{AD} = \ell_{DA} = 2.6$ Å, $\ell_{AA} = 3.5$ Å, and $\varphi_{AAD} = \varphi_{DAA} = 30^\circ$. Beside methanol [17], such geometric criteria have been applied to a variety of H-bonding fluids, in particular water [19, 20, 21], methanol-water mixtures [22], and ethanol [18, 23, 24], including previous molecular simulations with an unreleased development version of the *ms2* program [18]. H bonds in sorbate-sorbent interactions can be treated analogously [25]. For hydrogen fluoride, a simple distance-based criterion can be used [26], which is also covered by the present implementation.

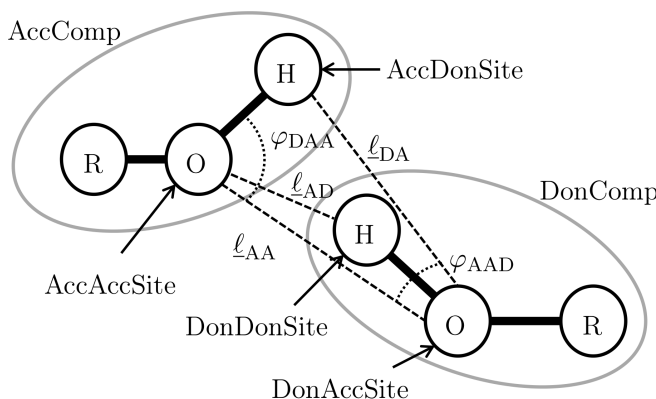


Figure 1: Class of H-bonding criteria implemented in the present release version of *ms2*.

- [1] D. Frenkel, B. Smit, *Understanding Molecular Simulation: From Algorithms to Applications*, Academic Press, Elsevier, San Diego, CA, 2002.
- [2] R. Lustig, Microcanonical monte carlo simulation of thermodynamic properties, *J. Chem. Phys.* 109 (20) (1998) 8816–8828. doi:http://dx.doi.org/10.1063/1.477552.
- [3] T. Kristóf, G. Rutkai, L. Merényi, J. Liszi, Molecular simulation of the joulethomson inversion curve of hydrogen sulphide, *Mol. Phys.* 103 (4) (2005) 537–545. doi:http://dx.doi.org/10.1080/00268970413331319263.
- [4] H. C. Andersen, Molecular dynamics simulations at constant pressure and/or temperature, *J. Chem. Phys.* 72 (1980) 2384–2393. doi:http://dx.doi.org/10.1063/1.439486.

- [5] R. Span, Multiparameter Equations of State: An Accurate Source of Thermodynamic Property Data, Springer Verlag, Berlin, 2000.
- [6] R. Lustig, Direct molecular nvt simulation of the isobaric heat capacity, speed of sound and joulethomson coefficient, *Mol. Sim.* 37 (06) (2011) 457–465. doi:http://dx.doi.org/10.1080/08927022.2011.552244.
- [7] R. Lustig, Statistical analogues for fundamental equation of state derivatives, *Mol. Phys.* 110 (24) (2012) 3041–3052. doi:http://dx.doi.org/10.1080/00268976.2012.695032.
- [8] C. W. Glass, S. Reiser, G. Rutkai, S. Deublein, A. Koster, G. Guevara-Carrion, A. Wafai, M. Horsch, M. Bernreuther, T. Windmann, H. Hasse, J. Vrabec, : A molecular simulation tool for thermodynamic properties, new version release, *Comput. Phys. Commun.* 185 (12) (2014) 3302–3306. doi:http://dx.doi.org/10.1016/j.cpc.2014.07.012.
- [9] H. Flyvbjerg, H. G. Petersen, Error estimates on averages of correlated data, *J. Chem. Phys.* 91 (1) (1989) 461–466. doi:http://dx.doi.org/10.1063/1.457480.
- [10] B. Widom, Some topics in the theory of fluids, *J. Chem. Phys.* 39 (11) (1963) 2808–2812. doi:http://dx.doi.org/10.1063/1.1734110.
- [11] T. Kristóf, G. Rutkai, Chemical potential calculations by thermodynamic integration with separation shifting in adaptive sampling monte carlo simulations, *Chem. Phys. Lett.* 445 (1-3) (2007) 74–78. doi:http://dx.doi.org/10.1016/j.cplett.2007.07.054.
- [12] R. Krishna, J. M. van Baten, The darken relation for multicomponent diffusion in liquid mixtures of linear alkanes: an investigation using molecular dynamics (md) simulations, *Ind. Eng. Chem. Res.* 44 (17) (2005) 6939–6947. doi:http://dx.doi.org/10.1021/ie050146c.
- [13] F. H. Stillinger, Water revisited, *Science* 209 (1980) 451–457.
- [14] R. H. Henchman, S. J. Irudayam, Topological hydrogen-bond definition to characterize the structure and dynamics of liquid water, *Journal of Physical Chemistry B* 114 (2010) 16792–16810.
- [15] E. Arunan, G. R. Desiraju, R. A. Klein, J. Sadlej, S. Scheiner, I. Alkorta, D. C. Clary, R. H. Crabtree, J. J. Dannenberg, P. Hobza, H. G. Kjaergaard, A. C. Legon, B. Mennucci, D. J. Nesbitt, Definition of the hydrogen bond (IUPAC recommendations 2011), *Pure and Applied Chemistry* 83 (2011) 1637–1641.
- [16] J. Thar, B. Kirchner, Hydrogen bond detection, *Journal of Physical Chemistry A* 110 (2006) 4229–4237.
- [17] M. Haughney, M. Ferrario, I. R. McDonald, Molecular-dynamics simulation of liquid methanol, *Journal of Physical Chemistry* 91 (1987) 4934–4940.

- [18] S. Reiser, N. McCann, M. Horsch, H. Hasse, Hydrogen bonding of ethanol in supercritical mixtures with CO₂ by ¹H NMR spectroscopy and molecular simulation, *Journal of Supercritical Fluids* 68 (2012) 94–103.
- [19] A. Luzar, D. Chandler, Structure and hydrogen bond dynamics of water – dimethyl sulfoxide mixtures by computer simulations, *Journal of Chemical Physics* 98 (1993) 8160–8173.
- [20] S. Chowdhuri, A. Chandra, Dynamics of halide ion – water hydrogen bonds in aqueous solutions: Dependence on ion size and temperature, *Journal of Physical Chemistry B* 110 (2006) 9674–9680.
- [21] A. Luzar, D. Chandler, Hydrogen-bond kinetics in liquid water, *Nature* 379 (1996) 55–57.
- [22] J. R. Choudhuri, A. Chandra, Hydrogen bonded structure, polarity, molecular motion and frequency fluctuations at liquid-vapor interface of a water-methanol mixture: An ab initio molecular dynamics study, *Journal of Chemical Physics* 141 (2014) 134703.
- [23] L. Saiz, J. A. Padró, E. Guàrdia, Structure and dynamics of liquid ethanol, *Journal of Physical Chemistry B* 101 (1997) 78–86.
- [24] W. Xu, J. Yang, Y. Hu, Microscopic structure and interaction analysis for supercritical carbon dioxide – ethanol mixtures: A Monte Carlo simulation study, *Journal of Physical Chemistry B* 113 (14) (2009) 4781–4789.
- [25] D. Bhandary, S. Khan, J. K. Singh, Structure and dynamics of n-alkohol monolayers on a mica surface, *Journal of Physical Chemistry C* 118 (2014) 6809–6819.
- [26] M. Kreitmeir, H. Bertagnolli, J. J. Mortensen, M. Parrinello, Ab initio molecular dynamics simulation of hydrogen fluoride at several thermodynamic states, *Journal of Chemical Physics* 118 (2003) 3639–3645.